

Future Interns – Data Science & Analytics Internship

Task 2: Customer Churn Analysis Using Power BI

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Internship: Future Interns – Data Science & Analytics

Task: Customer Retention & Churn Analysis

Tools Used: Python (Pandas, Matplotlib), Jupyter Notebook, Power BI

Project Overview

The objective of this project was to analyze customer churn data and identify the major factors contributing to customer attrition. The dataset was cleaned using Python and visualized in Power BI to develop an interactive dashboard that provides meaningful business insights.

Data Preparation

The dataset was imported into Jupyter Notebook for preprocessing. TotalCharges was converted to a numeric data type, missing values were removed, duplicate records were checked, exploratory data analysis (EDA) was performed, and the cleaned dataset was exported for Power BI.

Dashboard Components

The dashboard includes KPI cards for Total Customers, Churned Customers, Active Customers, Churn Rate, and Average Monthly Charges. Visualizations include Contract Type vs Customer Churn, Internet Service vs Customer Churn, Payment Method vs Customer Churn, Gender vs Customer Churn, Senior Citizen vs Customer Churn, Monthly Charges Distribution, and Customer Tenure Distribution. Interactive slicers for Gender, Contract Type, and Internet Service enable dynamic filtering.

Key Findings

- Month-to-Month contract customers experienced the highest churn.
- Fiber Optic users showed a higher churn rate than other internet services.
- Customers using Electronic Check were more likely to churn.
- Customers with longer tenure had better retention.
- Higher monthly charges were associated with increased churn.

Conclusion

This project demonstrates how Python and Power BI can be combined to clean, analyze, and visualize customer data. The dashboard highlights customer segments with higher churn risk and provides actionable insights to support customer retention strategies and business decision-making.